Problem sheets

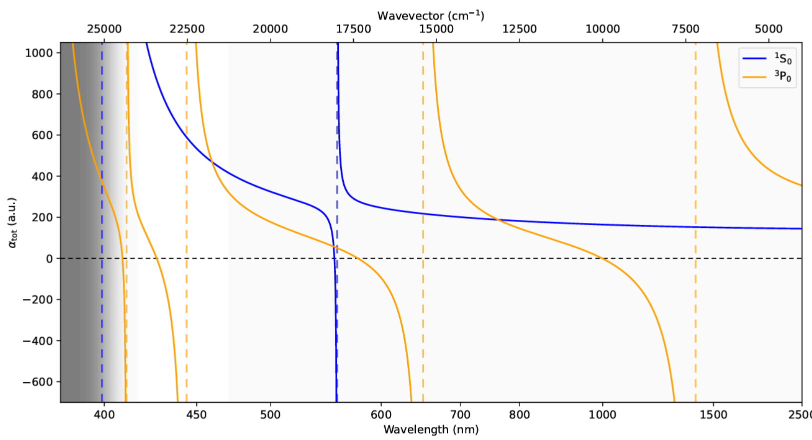
Quantum Technology Based on Atomic, Molecular, and Optical Systems (2025/ 09/30)

Email your Homework to chenchunchia@gmail.com
with title [HW - Atomic clock] **Your name -** Quantum Technology Based on Atomic, Molecular, and Optical Systems

Optical lattice Clocks:

(a) In an optical lattice clock, we use the “magic wavelength” to eliminate perturbations from the optical trapping potential. Discuss what systematic errors occur if the trapping laser wavelength deviates from the magic wavelength.

(b) In the polarizability plot of Yb ($^1S_0$ and $^3P_0$ states), identify the possible magic wavelengths in the range **400–2500 nm**. List the crossing points in ascending order of λ ($\lambda_1 < \lambda_2 < \lambda_3$ …).



(c) Experimental implementations typically require **500 mW–5 W** of optical power. For each magic wavelength identified in part (b), list a feasible laser source available on the market and justify your choice (considering power, tunability, and robustness).

(This is a the same problem you encountered when choosing between an iphone or Google pixel. I assume you have a reason)

(d) **Trap depth estimation**: The optical dipole trap depth is given by $U_{trap} \propto \frac{1}{2c\epsilon_0}\alpha I$, where α is the polarizability and I is the intensity. Assuming the same beam waist geometry, calculate the expected *power ratio* required to realize the same trap depth at two different magic wavelengths, e.g. λ_1 vs λ_3.

(e) So far we have discussed only *attractive* magic-wavelength traps. Is it possible to realize a **repulsive magic-wavelength trap**? Refer to Katori et al., *Phys. Rev. Lett. 102, 063002 (2009) (*https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.102.063002*)*.

(f) The ac Stark shift of a state *n* due to light of frequency ω can be written (second-order perturbation theory):

$$\Delta E_n(\omega) = \sum_{n'} \frac{|\langle n|d|n'\rangle|^2}{\hbar} \frac{\omega_{nn'}}{\omega_{nn'}^2 - \omega^2},$$

where the sum runs over dipole-allowed states *n′*, Γ is the decay rate, and $\omega_{nn'}$ is the transition frequency.

(a) Using NIST atomic spectra data, (Using NIST line data, (https://physics.nist.gov/PhysRefData/ASD/lines_form.html)

list five lowest-energy dipole-allowed transitions from the clock state $^3P_0$ in each of the following atoms:

- Hg (6s6p $^3P_0 \rightarrow {}^3P_1$ and $^3D_1$ series)
- Sr (5s5p $^3P_0 \rightarrow \ldots$)
- Yb (6s6p $^3P_0 \rightarrow \ldots$)

(b) Explain why the **Hg clock transition ($^1S_0$–$^3P_0$)** is less sensitive to **blackbody radiation shifts** than the Yb and Sr clock transitions.